# CAN THE SEA HOLD MORE $CO_2$?

Over the past decade the ocean took in 29% of the $CO_2$ we emitted. Since 2016 that uptake has stopped growing. Ocean alkalinity enhancement (OAE) might safely raise it. Nobody knows yet. This brief asks for the independent research that would find out.

**29%**

of the $CO_2$ people emitted over the past decade was taken up by the ocean

**2016**

the year that uptake stopped growing, at about 3.4 billion tonnes of carbon a year

**10,000+**

years carbon can stay dissolved as bicarbonate, if the added alkalinity persists

## WHAT IT IS

OAE adds alkaline material to surface seawater, either ground minerals such as olivine or a hydroxide made with electricity. The water then holds more carbon as stable bicarbonate and draws more $CO_2$ from the air. It speeds up a natural process. It is not ready to deploy.

## WHAT THE EVIDENCE SHOWS

- **One permitted sea trial found no harm in what it measured.** In the Gulf of Maine in August 2025, researchers found no significant difference in plankton, fish larvae or lobster larvae. The results are preliminary and not yet peer reviewed.
- **Much was not measured:** fish, marine mammals and fisheries.
- **A lab result fell short.** A 2022 study found olivine removed about five times less carbon than theory predicts.
- **Verification is still an open problem.** CarbonPlan puts confidence in verifying the electrochemical route at 3 out of 5. Project Drawdown lists OAE under Keep Watching.

## WHO PAYS SO FAR

- **Philanthropy.** The Carbon to Sea Initiative launched in 2023 with $50 million from foundations. It funds OAE research and also advocates for the field. It paid for the Gulf of Maine trial with ICONIQ Ocean Co-Lab.
- **Credit buyers.** In 2025 companies including Stripe, Google and Shopify agreed to pay $31.3 million for 115,211 tons of removal from one OAE company. That buys future removal, not research.
- **Public money.** US federal spending on ocean carbon removal research came to nearly $61.5 million across 2023 and 2024, by Carbon to Sea's count. We found no oil company among the named funders.

**The one permitted sea trial was privately funded. That is why the ask is for public money and independent review.**

## FIVE ASKS FOR YOUR DELEGATION

1. **Fund independent verification research.** An open, global standard for monitoring, reporting and verification, publicly funded and independently reviewed, in place before any credit counts toward a climate target.
2. **Set ecological guardrails.** Thresholds for pH, trace metals and biodiversity that pause a trial automatically, with independent oversight.
3. **Clarify governance.** Jurisdiction, liability and consent for ocean trials under UNCLOS and the London Protocol.
4. **Share decisions and benefits.** Coastal and Indigenous communities hold real decision power over trials.
5. **Keep emission cuts first.** OAE adds to rapid decarbonization and can never replace it.

## TWO MINUTES WITH YOUR DELEGATION? ASK

- Does our climate research budget fund independent work on verifying OAE, and who reviews it?
- Will we keep OAE credits out of national targets until an open, public standard exists?
- Who in our delegation follows ocean carbon removal, and will they raise research governance before COP32?

Scan for every source, the full case and a letter you can send to your country's climate focal point.

**alkalinityask.life**

Sources: Global Carbon Budget 2025 · National Academies, 2022 · Woods Hole Oceanographic Institution, February 2026 · MIT Technology Review, 2022 and 2023 · CarbonPlan CDR Verification Framework · Project Drawdown, 2025 · Frontier, 2025 · Carbon to Sea Initiative. Full references are on the page.

An independent brief, not affiliated with the organisations it cites. Figures as of 11 November 2026.